PREP 5 · LESSONS 9–10

I–V investigations and sensors

Investigating I–V characteristics (required practical) · thermistors, LDRs and sensing circuits

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Explain why the resistance of a filament lamp increases with current. [1]

2 State the function of a diode. [1]

3 Describe the total resistance of two resistors connected in parallel. [1]

Key ideas LESSONS 9–10

Show your working

4 A student finds the I–V characteristic of a lamp.

a Name the independent and dependent variables. [1]

b State why a variable resistor is included in the circuit. [1]

c Explain why the student reverses the connections to the lamp. [1]

5 As the temperature of a thermistor increases, its resistance **increases** | **decreases** | **stays the same** [1]

6 In the dark, the resistance of an LDR is **high** | **low** [1]

7 A thermistor is connected in series with a fixed resistor and a battery. The circuit switches on a fan when the room gets hot.

Explain what happens to the current in the thermistor as the room warms up. [2]